PETRA ZLATARIĆ-NOVOSEL

Machine Learning Engineer | NLP & Document AI

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PROFESSIONAL SUMMARY

Machine learning engineer with 5+ years of experience in applied machine learning, natural language processing and document AI. Experienced in taking document classification and NLP models into production, including model serving, production monitoring and cloud deployment on Azure. Strong Python background with PyTorch, Hugging Face Transformers and MLflow. Currently pursuing a PhD focused on named entity recognition in Croatian legal texts.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | Slavonica Document AI d.o.o. | Osijek *03/2023–Present*

- Developed and deployed an incoming-document classifier for public administration documents, achieving 94% accuracy and 0.91 macro F1 across 23 classes; the service processes approximately 15,000 documents per day.
- Fine-tuned XLM-R and BERTić transformer models for classification and named entity recognition in Croatian administrative documents.
- Delivered models as FastAPI services in Docker on Azure App Service, using Azure ML and MLflow for experiment tracking and model registry.
- Established weekly production monitoring for input text length, vocabulary and predicted class distribution, with alerts when PSI exceeds 0.2. Monitoring identified an OCR change that reduced accuracy by 9 percentage points; the issue was resolved within a week.
- Built an annotated dataset of 4,000 Croatian administrative documents, supported by annotation guidelines and two annotators; achieved inter-annotator Cohen's kappa of 0.87.
- Mentor two students during their professional internships.

Data Scientist | Baranjska kreditna banka d.d. | Zagreb (remote from Osijek) *09/2021–02/2023*

- Developed customer churn prediction models using gradient boosting and LightGBM for the sales team.
- Developed customer complaint topic classification, progressing from a TF-IDF and logistic regression baseline to fastText; the 11-topic classification supported weekly reporting to customer service management.
- Prepared SQL reports using the Oracle data warehouse.

Research Project Assistant | Faculty of Applied Informatics, University of Drava *10/2020–08/2021*

- Built a corpus of Croatian legal texts and annotated documents; wrote supporting Python scripts.

SELECTED PROJECTS

Retrieval-Augmented Question Answering Demo | Personal project *2025*

- Built a weekend prototype for question answering over Croatian laws using sentence embeddings, FAISS and an open-weight LLM accessed through an API.
- Checked outputs manually; the prototype was not used in production.

EDUCATION

PhD studies in Computer Science | University of Drava *2022–Present*
Research topic: Named entity recognition in Croatian legal texts | Expected 2027

MSc in Computer Science (Mag. ing. comp.) | University of Drava *2020*

CERTIFICATION

Microsoft Certified: Azure AI Engineer Associate | 2024

TECHNICAL SKILLS

Python, PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, fastText, spaCy, pandas, NumPy, SQL, MLflow, FastAPI, Docker, Azure ML, Azure App Service, Git, Linux, FAISS, sentence-transformers

TEACHING

Guest lecturer, NLP Summer School | 2025 | 90-minute lecture

PUBLICATION

Zlatarić-Novosel, P. et al. (2024). “Named entity recognition for Croatian administrative documents.” Workshop on Slavic Language Processing.

LANGUAGES

Croatian (native) | English (C1) | German (B2)